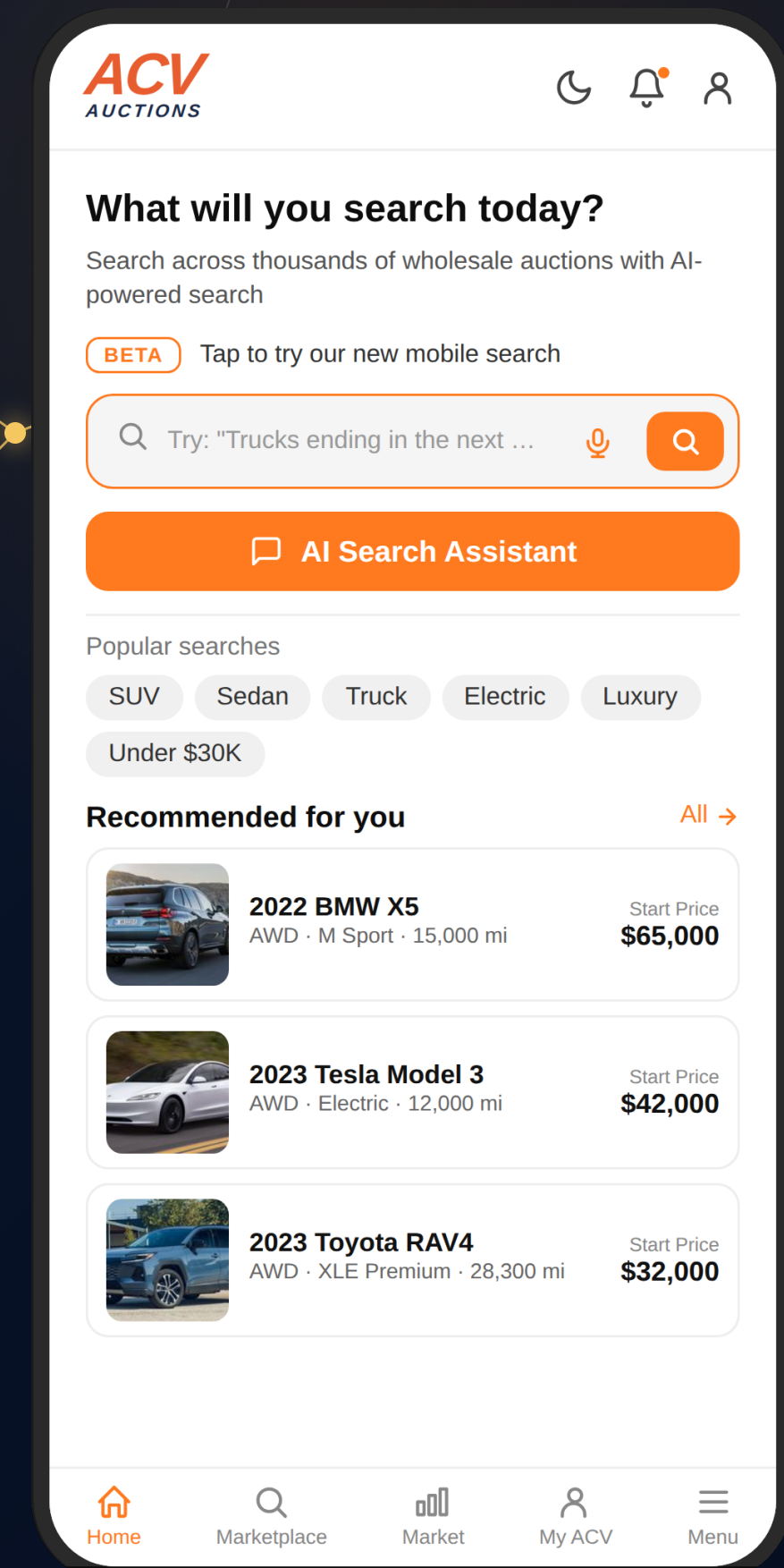


PRODUCT DESIGN CASE STUDY

ACV EASE

AI-powered search & discovery for wholesale vehicle auctions.

Margaux Kan · Senior Product Design Manager



TITLE & OUTCOME

Reimagining how dealers find their next car

ACV EASE is a mobile-first search experience for ACV's wholesale auction marketplace. Instead of stacking rigid filters, buyers ask for what they want in plain language, by voice or text — and an AI layer surfaces the strongest opportunities, explains why they matter, and guides each buyer from search to bid.

MY ROLE

Lead Product Designer

Research · IxD · UI · Prototyping

TIMELINE

14 weeks, 2025

PLATFORM

iOS & Android — ACV mobile app

TEAM

1 PM · 4 Engineers · 2 Data Scientists

STRATEGIC PROBLEM & RISK

Finding the right car was slow, manual, and overwhelming

Buyers run a high-volume business from their phones — between appraisals, on the lot, on the move. The old search made them work for it.

Filters, not language

Buyers had to translate what they wanted into rigid drop-downs and checkboxes, often missing inventory that didn't match the exact filter.

Thousands of live auctions

Relevant vehicles were scattered across constantly-changing auctions, hard to scan on a small screen with seconds to spare.

Good deals stayed buried

Nothing told a buyer which listing was actually a strong opportunity, so the best cars were easy to scroll right past.

MY ROLE & LEADERSHIP

Leading search from a strategic bet to launch

Lead product designer: research, interaction design, UI, and prototyping across the search-to-bid journey.

01 WHAT I LED

- Framed AI-powered search as a growth bet, not a UI refresh.
- Defined the vision for conversational, explainable discovery.
- Aligned Product, Engineering, and Data Science around buyer intent.
- Owned the search experience from discovery through beta launch.

02 HOW I LED

- Championed a mobile-first experience built for one hand on the lot.
- Partnered with Data Science on ranking and plain-language reasons.
- Led 3 design reviews; balanced speed against buyer trust.
- Used field evidence to resolve trade-offs across teams.

DESIGN PROCESS

A structured path from intent to confident bid

Five phases — pushing design upstream to shape the search strategy, not just the UI.

01

Understand

Studied how dealers search between appraisals and on the move.

02

Frame

Mapped the search-to-bid journey and a buyer-intent model.

03

Explore

Co-designed conversational and voice search patterns.

04

Validate

Prototyped & tested natural-language and voice search with dealers.

05

Deliver

Shipped a beta with Product, Engineering & Data Science.

LEADERSHIP LENS

Embedded design early to influence the ranking and intent strategy, not just the screens.

RESEARCH

Grounding AI search in how buyers actually search

Research led by me, with Product and Data Science embedded as partners.

METHODS

- Interviews & ride-alongs with 10 dealers across the U.S.
- Observed live search sessions on the legacy filter experience.
- Audited the old search; usability-tested NL & voice prototypes.

HOW RESEARCH WAS RUN

- Research plan led and owned by me.
- PMs embedded as co-researchers; Data Science on intent signals.
- Findings shared across Product, Engineering & Data Science.

GOAL *Learn where filter search broke down, and what a “good result” means to a buyer with seconds to decide.*

USER JOURNEY

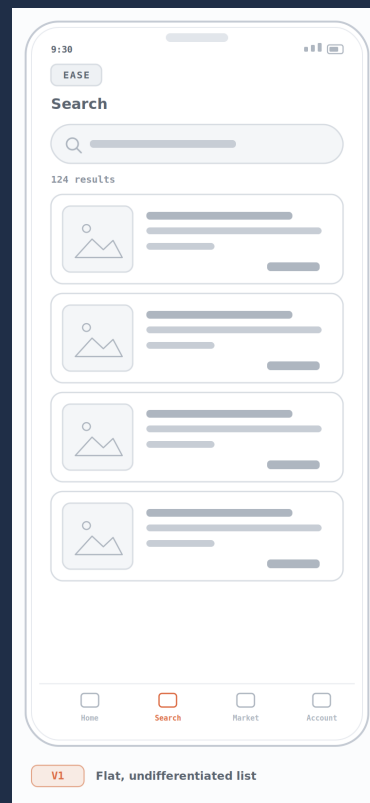
From first need to confident bid

	DISCOVER		EVALUATE	COMMIT	
STAGE	1. Spot a need	2. Search	3. Sift results	4. Size up the car	5. Decide & bid
DOING	On the lot	Voice / text	Results list	Vehicle page	Bid + assistant
FEELING	😊 Hopeful	😞 Friction	😓 Overwhelmed	😟 Unsure	😰 Anxious
PAIN POINT	<i>Restocking is reactive, time-boxed.</i>	<i>Wants don't fit rigid dropdowns.</i>	<i>Strong cars buried in the noise.</i>	<i>No read on whether the deal is good.</i>	<i>Commits on gut, low confidence.</i>
EASE RESPONSE	Meets buyers on mobile, in the moment.	Plain-language & voice search.	Surfaces one ranked top pick + why.	AI summary, bid score, success signal.	Confident bid; assistant one tap away.

INSIGHTS & ITERATIONS

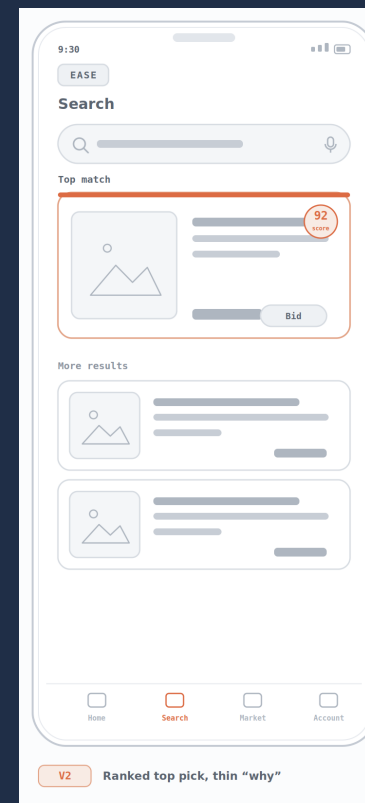
What we heard, and how the design evolved

Three insights reshaped the experience from a list into a recommendation.



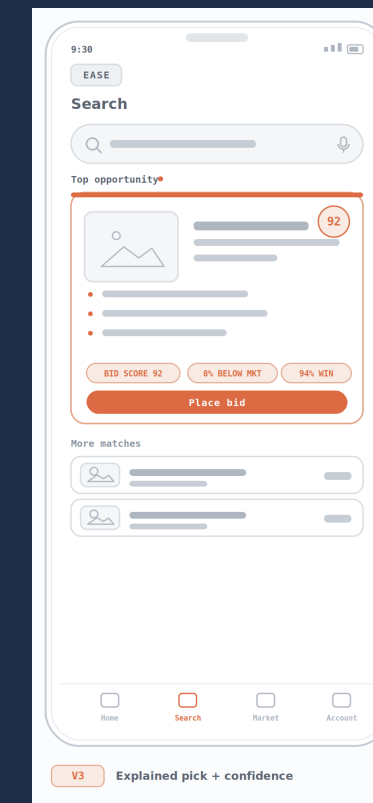
01 V1 — flat list

Buyers think in plain language, not filter taxonomies.



02 V2 — ranked pick

One strong, explained pick beats a long list of options.



03 V3 — explained + confidence

Trust comes from the why: reasons plus a confidence signal.

KEY LEARNING *Leading with one explained opportunity, plus plain-language and voice input, cut search friction without hiding inventory.*

From filtering a catalog to surfacing the best opportunity

PRIMARY PERSONA

High-Volume Dealer

- Mobile-first; buys between appraisals and on the lot.
- Optimizes for speed, margin, and the next profitable car.
- Low tolerance for friction; high need for trust.

STRATEGIC REFRAME

We shifted from helping buyers filter inventory to recommending the strongest opportunity — and explaining why it's worth a bid.

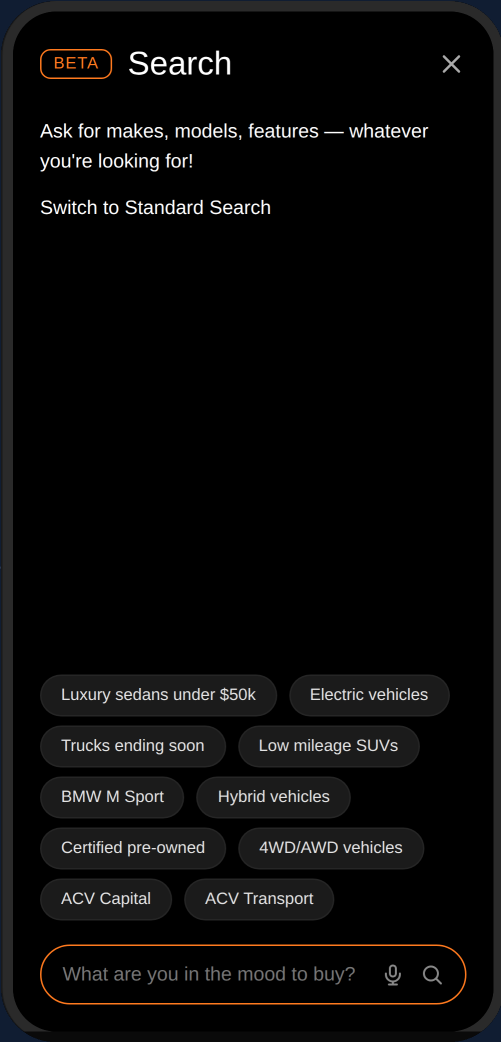
“EASE surfaces the one car I'd have missed. It's like having a co-buyer in my pocket.”

Jason M., Independent Dealer — Texas

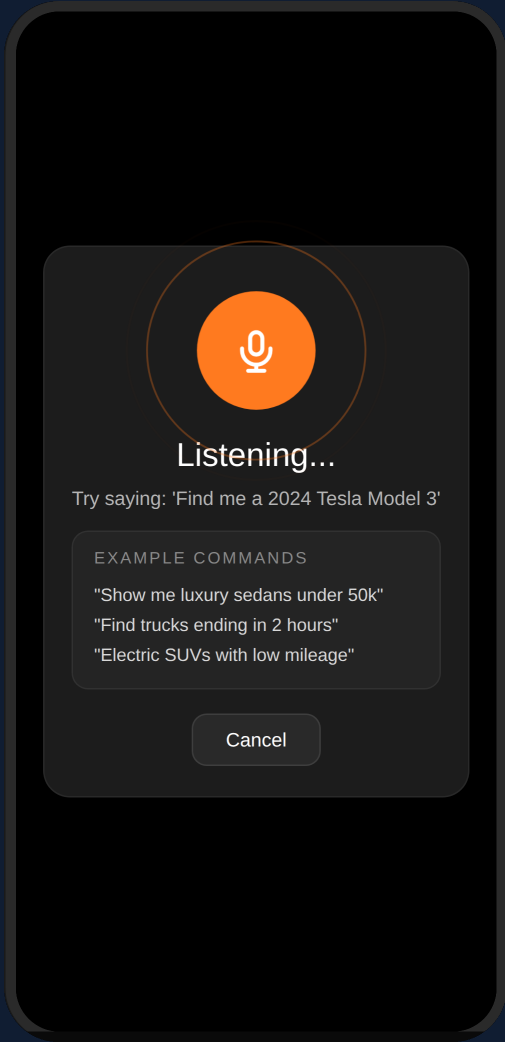
THE SOLUTION

One intelligent flow, end to end

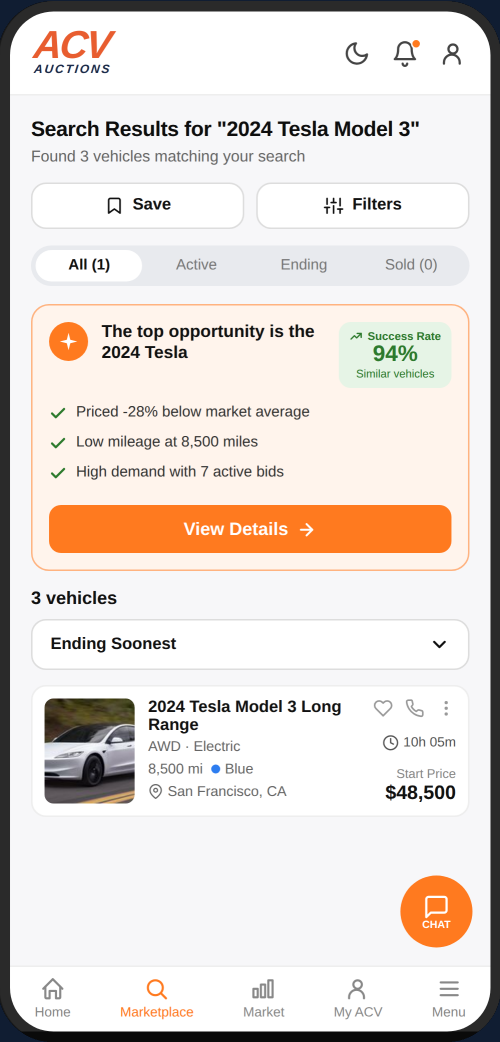
From a spoken request to a confident bid, the experience stays in one continuous, guided thread.



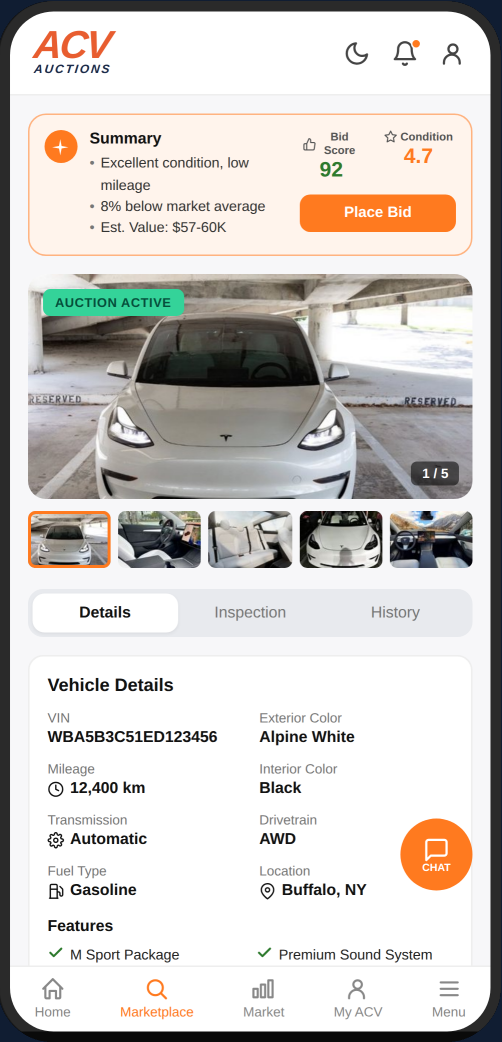
1Ask



2Understand



3Surface



4Decide

IMPACT

Early signals

Illustrative results from beta — to be replaced with measured launch data.

USER IMPACT

40%

Time to first relevant result
vs. the legacy filter search

4.3 / 5

Beta buyer satisfaction with the new search
n=42 dealers surveyed

BUSINESS IMPACT

+12 pts

Lift in search → vehicle-detail view
beta cohort vs. control

+6%

Bid conversion rate
beta cohort vs. control